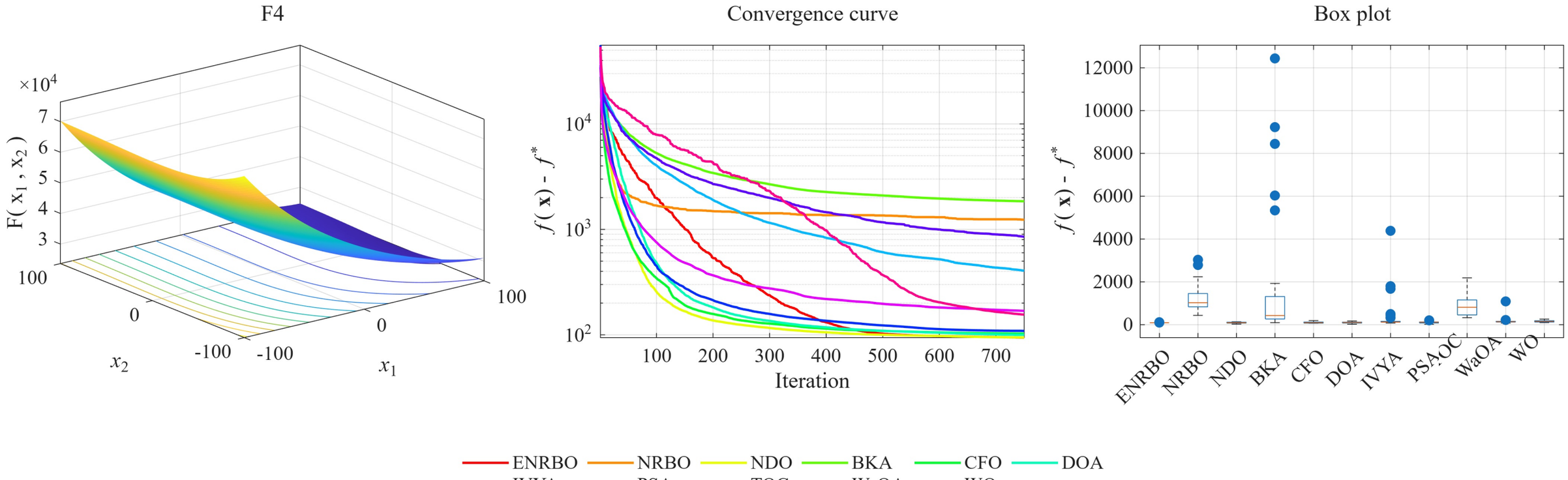

F4
×10^4
F(x1 , x2)
7
6
5
4
3
100
0
-100
x2
-100
0
100
x1
Convergence curve
f(x) - f*
10^4
10^3
10^2
100
200
300
400
500
600
700
Iteration
Box plot
f(x) - f*
12000
10000
8000
6000
4000
2000
0
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
OC
WaOA
WO
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
TOC
WaOA
WO